

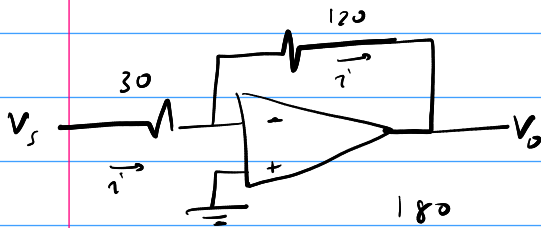
Sheet 1

2201377

Mohammed Hany

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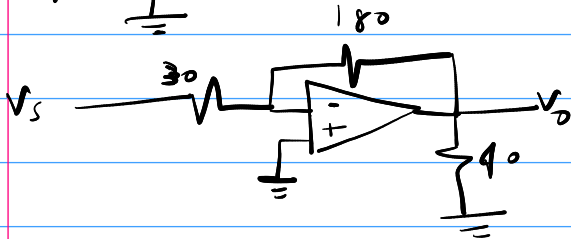
a)



$$\frac{V_s}{30} = \frac{-V_o}{120} \rightarrow \frac{V_o}{V_s} = -4$$

$$R_{in} = 30k\Omega$$

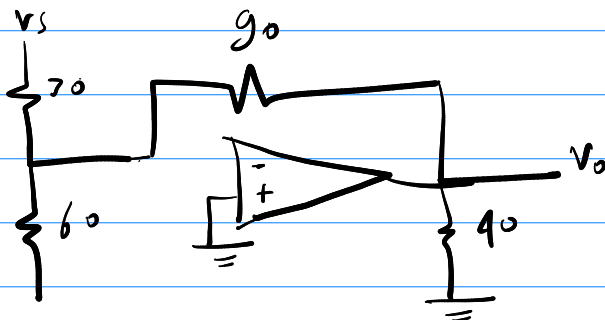
b)



$$\frac{V_s}{30} = \frac{-V_o}{180} \rightarrow \frac{V_o}{V_s} = -6$$

$$R_{in} = 30k\Omega$$

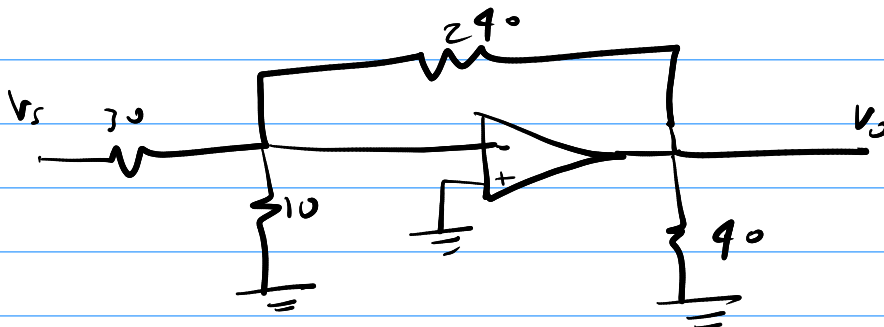
c)



$$\frac{V_s}{70} = \frac{-V_o}{90} \rightarrow \frac{V_o}{V_s} = -3$$

$$R_{in} = 30k\Omega$$

d)

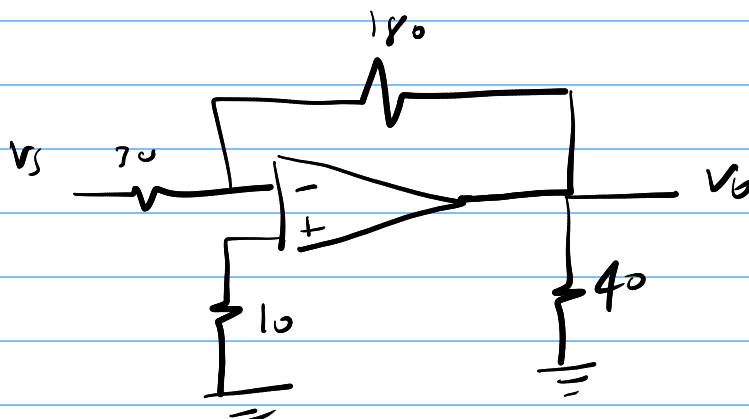


$$\frac{V_s}{70} = \frac{-V_o}{290}$$

$$\rightarrow \frac{V_o}{V_s} = -8$$

$$R_{in} = 30k\Omega$$

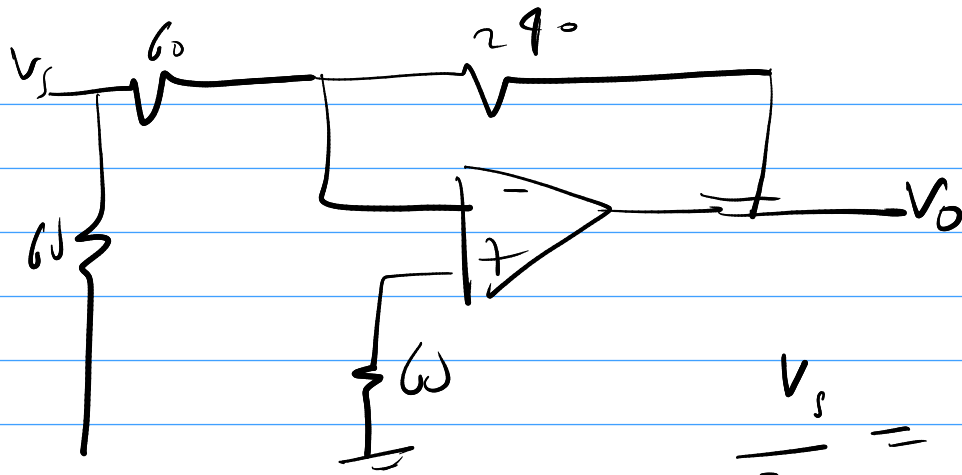
e)



$$\frac{V_s}{70} = \frac{-V_o}{180} \rightarrow \frac{V_o}{V_s} = -6$$

$$R_{in} = 30k\Omega$$

(f)



$$\frac{V_s}{30} = - \frac{V_o}{30}$$